

A Theoretical Deep Dive into Generative Adversarial Networks

From Provided Analysis

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Part I

Practical Implementation and Loss Functions

1 The Non-Saturating Generator Loss

In the standard Generative Adversarial Network (GAN) framework, the generator's objective is to produce samples that are indistinguishable from real data. A common issue with the original minimax loss function for the generator, $\min_G \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))]$, is that it can lead to vanishing gradients, especially early in training when the discriminator can easily reject generated samples (i.e., $D(G(z)) \approx 0$). When this happens, the gradient of the loss with respect to the generator's parameters becomes very small, stalling the learning process.

To address this, an alternative, widely-used objective is the "non-saturating" generator loss. Instead of minimizing the log-probability of the discriminator being correct, the generator aims to *maximize* the log-probability of the discriminator being incorrect. This is equivalent to minimizing the negative of that log-probability.

The implementation of this loss is straightforward. Assuming `disc_out_for_g` is a tensor of discriminator output:

The objective is to maximize $\log(D(G(z)))$, which is equivalent to minimizing $-\log(D(G(z)))$.

Add a small epsilon for numerical stability if `disc_out_for_g` can be exactly 0, though sigmoid outputs are typically $0 < \text{epsilon} = 1e - 8$. `g_loss = -torch.mean(torch.log(disc_out_for_g + epsilon))`

`g_loss = -torch.mean(torch.log(disc_out_for_g))` It's crucial that `disc_out_for_g` represents $D_\phi(G_\theta(z))$, i.e., the discriminator's probability output (after the sigmoid activation) for the generated samples. The `torch.log` function computes the natural logarithm, and `torch.mean` averages the loss over the mini-batch.

The change from minimizing $\log(1 - D(G(z)))$ to maximizing $\log D(G(z))$ (or minimizing $-\log D(G(z))$) fundamentally alters the gradient landscape for the generator. While both objectives ideally lead to the generator producing realistic samples (where $D(G(z))$ is high), the non-saturating version ensures that the learning signal remains strong even when the generator is far from this ideal state.

Table 1: Minimax vs. Non-Saturating Generator Loss Comparison

Feature	Minimax Loss ($L_{G_{\text{minimax}}}$)	Non-Saturating Loss ($L_{G_{\text{non-sat}}}$)
Mathematical Definition	$\mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))]$	$-\mathbb{E}_{z \sim p_z(z)} [\log(D(G(z)))]$
Conceptual Goal	Minimize probability of fakes being caught by discriminator.	Maximize probability of fakes being classified as real by discriminator.
Gradient Behavior (when $D(G(z)) \approx 0$)	Vanishing gradients (weak learning signal for G).	Strong gradients (robust learning signal for G).
Primary Advantage	Original theoretical formulation from the GAN paper.	Provides stronger, more consistent gradients early in training.
Common Issue Addressed	(This loss <i>has</i> the vanishing gradient issue)	Addresses the vanishing gradient problem of the minimax loss for G.

Part II

Theoretical Deep Dive into GANs and Divergence Measures

Beyond the practical aspects of training, GANs possess a rich theoretical foundation that connects their objective functions to fundamental concepts in probability and information theory.

2 Problem 2.1: Characterizing the Optimal Discriminator

A key step in understanding GAN theory is to determine the behavior of an optimal discriminator, D^* , given a fixed generator, G .

Problem 2.1. *Given a fixed generator G , find the optimal discriminator $D^*(x)$ that maximizes the value function $V(D, G)$.*

2.1 The Discriminator's Objective Revisited

The discriminator D_ϕ aims to maximize its objective function, which, as defined in the original GAN paper, is:

$$\max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}(x)}[\log D(x)] + \mathbb{E}_{z \sim p_z(z)}[\log(1 - D(G(z)))]$$

Letting $p_\theta(x)$ denote the distribution of samples generated by $G_\theta(z)$, this can be expressed as an integral over the data space $\mathcal{X}$:

$$V(D, G) = \int_{\mathcal{X}} (p_{\text{data}}(x) \log D(x) + p_\theta(x) \log(1 - D(x))) dx$$

To find the optimal discriminator $D^*(x)$ that maximizes this value function for a fixed generator (and thus fixed $p_{\text{data}}(x)$ and $p_\theta(x)$), we can maximize the integrand for each point x independently.

Solution 2.1.1. For a fixed x , let $a = p_{\text{data}}(x)$ and $b = p_\theta(x)$. We want to find the value $t = D(x) \in (0, 1)$ that maximizes the expression $g(t) = a \log t + b \log(1 - t)$.

We take the derivative of $g(t)$ with respect to t :

$$g'(t) = \frac{a}{t} - \frac{b}{1-t}$$

Setting $g'(t) = 0$ to find critical points:

$$\begin{aligned} \frac{a}{t} &= \frac{b}{1-t} \\ a(1-t) &= bt \\ a - at &= bt \\ a &= at + bt \\ a &= (a+b)t \\ t &= \frac{a}{a+b} \end{aligned}$$

To confirm this is a maximum, we examine the second derivative:

$$g''(t) = -\frac{a}{t^2} - \frac{b}{(1-t)^2}$$

Since $a = p_{\text{data}}(x) \geq 0$, $b = p_\theta(x) \geq 0$, and $t \in (0, 1)$, we have $g''(t) < 0$ (assuming a, b are not both zero). Thus, $t = \frac{a}{a+b}$ is indeed a maximum.

Substituting back $a = p_{\text{data}}(x)$ and $b = p_{\theta}(x)$, the optimal discriminator $D^*(x)$ is:

$$D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_{\theta}(x)} \quad (1)$$

This derivation is a cornerstone of GAN theory.

Interpretation 2.1.1. The form of the optimal discriminator $D^*(x)$ has a clear probabilistic interpretation: it is the probability that a given sample x belongs to the true data distribution $p_{\text{data}}(x)$, given that it originated from the mixture distribution formed by $p_{\text{data}}(x)$ and $p_{\theta}(x)$.

- If $p_{\text{data}}(x)$ is high and $p_{\theta}(x)$ is low (i.e., x is very likely real and unlikely generated), $D^*(x) \rightarrow 1$.
- If $p_{\theta}(x)$ is high and $p_{\text{data}}(x)$ is low (i.e., x is very likely generated and unlikely real), $D^*(x) \rightarrow 0$.
- Crucially, if the generator perfectly mimics the data distribution, such that $p_{\text{data}}(x) = p_{\theta}(x)$ for all x , then $D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_{\text{data}}(x)} = \frac{1}{2}$. In this scenario, the discriminator is maximally confused and cannot distinguish real from generated samples better than random guessing.

The significance of $D^*(x)$ extends beyond this interpretation. Its explicit form allows the original GAN objective $V(D^*, G)$ to be related to a statistical divergence measure between $p_{\text{data}}(x)$ and $p_{\theta}(x)$, namely the Jensen-Shannon divergence.

3 Problem 2.2: Discriminator Logits as Log-Likelihood Ratios

The output of the discriminator $D_{\phi}(x)$ is typically obtained by passing a "logit" value $h_{\phi}(x)$ through a sigmoid function. Understanding the meaning of these logits for an optimal discriminator provides further insight.

Problem 3.1. Show that if the discriminator is optimal ($D_{\phi} = D^*$), then its logits $h_{\phi}(x)$ are equal to the log-likelihood ratio, i.e., $h_{\phi}(x) = \log\left(\frac{p_{\text{data}}(x)}{p_{\theta}(x)}\right)$.

3.1 The Role of Logits and the Sigmoid Activation

The discriminator $D_{\phi}(x)$ is defined as $D_{\phi}(x) = \sigma(h_{\phi}(x))$, where $h_{\phi}(x)$ is the activation of the layer just before the final sigmoid function $\sigma(s) = 1/(1+e^{-s})$. The values $h_{\phi}(x)$ are referred to as the discriminator's logits.

Solution 3.1.1. We have the optimal discriminator from Equation 1:

$$D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_{\theta}(x)}$$

And we know that for an optimal discriminator, its output must be equal to this value, so $D^*(x) =$

$\sigma(h_\phi(x)) = \frac{1}{1+e^{-h_\phi(x)}}$. Equating these two expressions for $D^*(x)$:

$$\begin{aligned}
\frac{1}{1+e^{-h_\phi(x)}} &= \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_\theta(x)} \\
1+e^{-h_\phi(x)} &= \frac{p_{\text{data}}(x) + p_\theta(x)}{p_{\text{data}}(x)} \\
1+e^{-h_\phi(x)} &= 1 + \frac{p_\theta(x)}{p_{\text{data}}(x)} \\
e^{-h_\phi(x)} &= \frac{p_\theta(x)}{p_{\text{data}}(x)} \\
\ln(e^{-h_\phi(x)}) &= \ln\left(\frac{p_\theta(x)}{p_{\text{data}}(x)}\right) \\
-h_\phi(x) &= \log\left(\frac{p_\theta(x)}{p_{\text{data}}(x)}\right) \\
h_\phi(x) &= -\log\left(\frac{p_\theta(x)}{p_{\text{data}}(x)}\right) = \log\left(\left(\frac{p_\theta(x)}{p_{\text{data}}(x)}\right)^{-1}\right) \\
h_\phi(x) &= \log\left(\frac{p_{\text{data}}(x)}{p_\theta(x)}\right)
\end{aligned}$$

This shows that the logits of the optimal discriminator indeed estimate the log of the likelihood ratio of x under the true data distribution compared to the model's (generator's) distribution.

Interpretation 3.1.1. This result is quite revealing: the logits of an optimal GAN discriminator are not just arbitrary numbers but directly correspond to the log-likelihood ratio $\log(p_{\text{data}}(x)/p_\theta(x))$. This quantity is fundamental in statistical decision theory (e.g., the Neyman-Pearson lemma) and quantifies the evidence provided by sample x in favor of it originating from $p_{\text{data}}(x)$ versus $p_\theta(x)$. This implies that the discriminator, in its internal representation, is effectively computing a statistically meaningful measure of how much more (or less) likely a sample is to be real than generated, on a logarithmic scale.

4 Problem 2.3: A Specific Generator Loss and KL-Divergence

Different choices for the generator's loss function can lead to the optimization of different statistical divergences when an optimal discriminator is assumed.

Definition 4.1 (Kullback-Leibler (KL) Divergence). The KL divergence, or relative entropy, measures how one probability distribution P differs from a reference distribution Q . For continuous distributions, it is defined as:

$$\text{KL}(P \parallel Q) = \int P(x) \log\left(\frac{P(x)}{Q(x)}\right) dx = \mathbb{E}_{x \sim P(x)} \left[\log \frac{P(x)}{Q(x)} \right]$$

KL divergence is non-negative ($\text{KL}(P \parallel Q) \geq 0$) and asymmetric ($\text{KL}(P \parallel Q) \neq \text{KL}(Q \parallel P)$).

Problem 4.1. Consider the generator loss $L_G(\theta; \phi) = \mathbb{E}_{x \sim p_\theta(x)} \left[\log \frac{1-D_\phi(x)}{D_\phi(x)} \right]$. Show that if the discriminator is optimal ($D_\phi = D^*$), then minimizing this loss is equivalent to minimizing $\text{KL}(p_\theta(x) \parallel p_{\text{data}}(x))$.

Solution 4.1.1. We assume the discriminator is optimal, so we can use $D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_\theta(x)}$. First, let's find an expression for $1 - D^*(x)$:

$$1 - D^*(x) = 1 - \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_\theta(x)} = \frac{(p_{\text{data}}(x) + p_\theta(x)) - p_{\text{data}}(x)}{p_{\text{data}}(x) + p_\theta(x)} = \frac{p_\theta(x)}{p_{\text{data}}(x) + p_\theta(x)}$$

Now, we can compute the ratio inside the logarithm of the loss function:

$$\frac{1 - D^*(x)}{D^*(x)} = \frac{\frac{p_\theta(x)}{p_{\text{data}}(x) + p_\theta(x)}}{\frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_\theta(x)}} = \frac{p_\theta(x)}{p_{\text{data}}(x)}$$

Substituting this back into the generator's loss function:

$$\begin{aligned} L_G(\theta; \phi^*) &= \mathbb{E}_{x \sim p_\theta(x)} \left[\log \left(\frac{1 - D^*(x)}{D^*(x)} \right) \right] \\ &= \mathbb{E}_{x \sim p_\theta(x)} \left[\log \left(\frac{p_\theta(x)}{p_{\text{data}}(x)} \right) \right] \end{aligned}$$

By definition, this is exactly the Kullback-Leibler divergence $\text{KL}(p_\theta(x) \parallel p_{\text{data}}(x))$.

Interpretation 4.1.1. This result demonstrates that this particular generator loss, when coupled with an optimal discriminator, leads the generator to explicitly minimize $\text{KL}(p_\theta \parallel p_{\text{data}})$. This is often referred to as the "reverse KL" divergence. This objective is known for its **mode-seeking** behavior. The generator p_θ is heavily penalized if it produces samples x for which $p_\theta(x) > 0$ but $p_{\text{data}}(x)$ is very low. To avoid this, p_θ tends to focus its probability mass on regions where p_{data} has high density. While this can lead to high-quality samples, it also carries the risk of **mode collapse**, where the generator learns to produce samples from only a subset of the modes present in p_{data} .

5 Problem 2.4: Comparing VAEs and GANs

Variational Autoencoders (VAEs) and GANs are two prominent families of deep generative models. Their underlying objectives lead to distinct characteristics.

Problem 5.1. A VAE's decoder is trained by maximizing the Evidence Lower Bound (ELBO), which is equivalent to minimizing the negative log-likelihood (NLL) of the data, $-\mathbb{E}_{x \sim p_{\text{data}}(x)}[\log p_\theta(x)]$.

1. Show that minimizing this NLL is equivalent to minimizing $\text{KL}(p_{\text{data}}(x) \parallel p_\theta(x))$.
2. Compare this objective to the GAN generator objective from Problem 2.3. Are they learning the same thing?

Solution 5.1.1 (Part 1: NLL as KL Divergence). We want to relate the NLL to the "forward" KL divergence, $\text{KL}(p_{\text{data}} \parallel p_\theta)$. By definition:

$$\text{KL}(p_{\text{data}}(x) \parallel p_\theta(x)) = \mathbb{E}_{x \sim p_{\text{data}}(x)} \left[\log \left(\frac{p_{\text{data}}(x)}{p_\theta(x)} \right) \right]$$

Using the property of logarithms, $\log(A/B) = \log A - \log B$:

$$\text{KL}(p_{\text{data}}(x) \parallel p_\theta(x)) = \mathbb{E}_{x \sim p_{\text{data}}(x)}[\log p_{\text{data}}(x)] - \mathbb{E}_{x \sim p_{\text{data}}(x)}[\log p_\theta(x)]$$

Rearranging this equation to solve for the NLL term:

$$-\mathbb{E}_{x \sim p_{\text{data}}(x)}[\log p_\theta(x)] = \text{KL}(p_{\text{data}}(x) \parallel p_\theta(x)) - \mathbb{E}_{x \sim p_{\text{data}}(x)}[\log p_{\text{data}}(x)]$$

The term $\mathbb{E}_{x \sim p_{\text{data}}(x)}[\log p_{\text{data}}(x)]$ is the negative entropy of the true data distribution, $-H(p_{\text{data}}(x))$. Since $p_{\text{data}}(x)$ is fixed and does not depend on the model parameters θ , this term is a constant with respect to the optimization of θ . Thus, we have:

$$\underbrace{-\mathbb{E}_{x \sim p_{\text{data}}(x)}[\log p_\theta(x)]}_{\text{NLL}} = \underbrace{\text{KL}(p_{\text{data}}(x) \parallel p_\theta(x))}_{\text{Forward KL}} + \underbrace{H(p_{\text{data}}(x))}_{\text{Constant w.r.t. } \theta}$$

This shows that minimizing the negative log-likelihood with respect to θ is equivalent to minimizing $\text{KL}(p_{\text{data}}(x) \parallel p_\theta(x))$.

Solution 5.1.2 (Part 2: Comparison). The question asks whether a VAE decoder and the GAN generator from Problem 2.3 are implicitly learning the same objective.

- A **VAE decoder**, by maximizing the ELBO, effectively minimizes the forward KL divergence, $\text{KL}(p_{\text{data}} \parallel p_\theta)$.
- The **GAN generator** from Problem 2.3, when paired with an optimal discriminator, minimizes the reverse KL divergence, $\text{KL}(p_\theta \parallel p_{\text{data}})$.

These two objectives, $\text{KL}(p_{\text{data}} \parallel p_{\theta})$ and $\text{KL}(p_{\theta} \parallel p_{\text{data}})$, are **not the same** due to the asymmetry of the KL divergence. They lead to different optimization behaviors:

- **Forward KL ($\text{KL}(p_{\text{data}} \parallel p_{\theta})$ - VAE-like):** This divergence is minimized when $p_{\theta}(x)$ is high wherever $p_{\text{data}}(x)$ is high. If $p_{\theta}(x) \rightarrow 0$ for some x where $p_{\text{data}}(x) > 0$, the term $\log(p_{\text{data}}(x)/p_{\theta}(x))$ becomes very large, heavily penalizing the model. To avoid this, p_{θ} tends to "cover" all regions where p_{data} has mass. This is known as **mode-covering** behavior. This can lead to blurry or averaged samples if the model is not expressive enough.
- **Reverse KL ($\text{KL}(p_{\theta} \parallel p_{\text{data}})$ - GAN-like):** This divergence is minimized when $p_{\theta}(x)$ is low wherever $p_{\text{data}}(x)$ is low. If $p_{\theta}(x) > 0$ for some x where $p_{\text{data}}(x) \approx 0$, the model is heavily penalized. To avoid this, p_{θ} tends to only place mass in regions where p_{data} has significant mass. This is known as **mode-seeking** behavior, which can result in sharp samples but may lead to mode collapse.

Therefore, a VAE decoder and the specific GAN generator from Problem 2.3 are **not** implicitly learning the same objective.

Table 2: VAE vs. GAN (Problem 2.3 Formulation) Objective Comparison

Aspect	VAE (Decoder Objective via ELBO)	GAN (Generator Objective from Problem 2.3)
Effective Divergence	Approximately $\text{KL}(p_{\text{data}}(x) \parallel p_{\theta}(x))$	$\text{KL}(p_{\theta}(x) \parallel p_{\text{data}}(x))$
Direction of KL-Divergence	Forward KL	Reverse KL
Primary Penalty Focus	Model assigns low probability to real data points ($p_{\theta}(x) \approx 0$ where $p_{\text{data}}(x) > 0$)	Model generates samples not in true data ($p_{\theta}(x) > 0$ where $p_{\text{data}}(x) \approx 0$)
Implied Behavior	Mode-covering	Mode-seeking
Typical Impact on Sample Quality	Can be less sharp, potentially blurry or averaged.	Can be sharp and high-quality.
Typical Impact on Sample Diversity	Tends to cover all modes of the data distribution.	May suffer from mode collapse (misses some modes).

Part III

Synthesis and Concluding Thoughts

6 Summary of Key Learnings and Implications

This exploration of Generative Adversarial Networks has traversed from the foundational mechanics of their adversarial training to the deeper theoretical underpinnings that govern their behavior. Several key themes emerge:

1. **The Critical Role of Loss Functions:** The choice of loss function is paramount. The initial minimax loss suffers from vanishing gradients, while the non-saturating loss provides a pragmatic solution. More advanced losses connect the GAN objective to specific statistical divergences.

2. The Optimal Discriminator as a Theoretical Tool: The derivation of $D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_{\theta}(x)}$ is a cornerstone of GAN theory. It allows the GAN objective to be re-expressed as minimizing a divergence and reveals that the optimal discriminator's logits estimate the log-likelihood ratio $\log(p_{\text{data}}(x)/p_{\theta}(x))$.
3. GANs and Divergence Minimization: Many GAN formulations can be understood as methods for minimizing some form of statistical divergence between probability distributions. The specific divergence (e.g., Jensen-Shannon, forward KL, reverse KL, Wasserstein distance) dictates the learning dynamics.
4. The Mode-Seeking vs. Mode-Covering Dichotomy: This distinction is crucial. Mode-seeking objectives (like reverse KL) tend to produce high-quality but potentially less diverse samples (risk of mode collapse). Mode-covering objectives (like forward KL) aim for diversity, sometimes at the expense of sample sharpness.
5. Ongoing Evolution of GANs: The challenges of GAN training have spurred vast research into alternative loss functions (e.g., Wasserstein GAN [2, 4]), architectural innovations, and regularization techniques to make GANs more robust and capable.

In conclusion, GANs represent a powerful and flexible framework for generative modeling. Harnessing this power requires a careful understanding of their theoretical properties, the nuances of their loss functions, and the practical challenges of their training. The connections to information theory and statistical divergence measures offer a unifying perspective, guiding the design of new and improved generative models.

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